

Detecting Abnormal Markets

Early Warning Systems with Deep & Classical Anomaly Detection

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Why this matters

Crashes are rare but decisive – 1929, 2000, 2008, 2020 ...

A handful of days drives a lifetime of returns.

Buy-and-hold S&P 500, 1927–2015:

Strategy	Growth of \$1
Stay invested	\$115.40
Miss the 10 worst days	\$362.01
Miss the 10 best days	\$38.28

Tail events cluster

The frequency and cross-asset correlation of simultaneous heavy losses keep rising.

Diversification fails exactly when you need it most – so we need a signal that fires *before* the drawdown compounds.

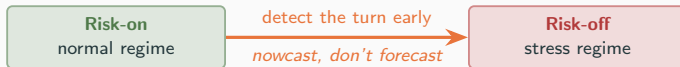
The transition we must catch

Risk-on

Confidence. Equities, credit, EM, commodities bid up. Volatility low, spreads tight, capital chases growth.

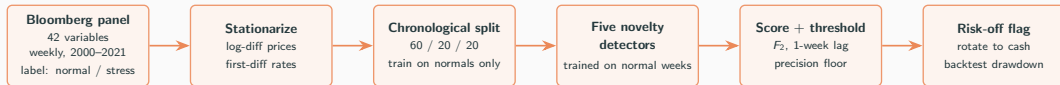
Risk-off

Fear. Flight to cash, short govies, gold, CHF, JPY. Volatility spikes, correlations surge – systemic risk.



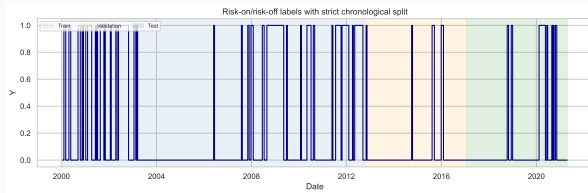
We do not predict *which* crash. We flag, in real time, that this week is “**not normal**” – a 2008-style and a 2020-style stress both qualify.

Pipeline: from raw prices to a risk-off flag

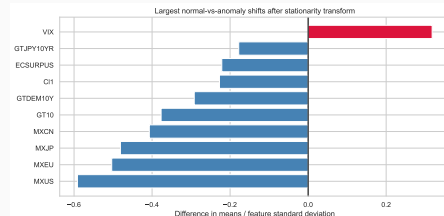


- Treat **normal markets** as the inlier distribution; train on normals only and flag high anomaly scores – no need to label crisis types ex ante.
- Deployment: a flag rotates into cash / short govies / gold. User: CIO, risk officer, wealth manager. **Goal = cut drawdown**, not chase alpha.

Data: stress clusters around drawdowns



US equity (MXUS) with stress weeks shaded



Largest standardized mean shifts, normal vs. stress

1110 weekly observations $\times$ 42 features; **21%** of weeks labelled abnormal. Stress weeks line up with equity selloffs and load on equity returns (MXUS, MXEU, MXJP) and the VIX – the signal is real and economically sensible.

Five detectors, two families

Train on **normal weeks only** $\Rightarrow$ anomaly score $\Rightarrow$ threshold $\Rightarrow$ risk-off flag

Classical / density

Isolation Forest (+ Top- K) – anomalies isolate in fewer random splits

COPOD – empirical-copula tail probabilities, asset-grouped sub-models

Deep generative

GAN – discriminator realism

1 – $D(x)$ & latent reconstruction

VAE – KL-regularised reconstruction error

LSTM-AE – 12-week window, recon + Mahalanobis

One shared objective (F_2) tunes every threshold $\Rightarrow$ comparable operating points across all five.

Honest evaluation: no look-ahead

Train · 60%	CV · 20%	Test · 20%
2000–2012 472 normal weeks	2012–2017 tune threshold	2017–2021 29 crises (13%)

Strict chronological split

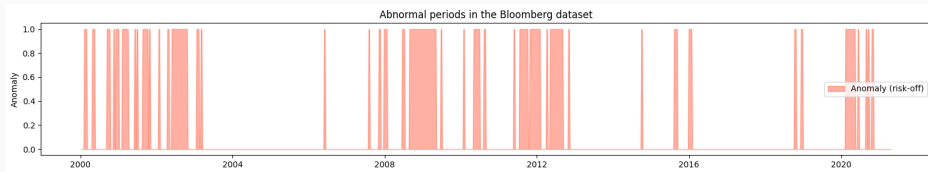
- Fit on the *past* only – faithful to live deployment.
- Threshold: all five maximise F_2 ($\beta = 2.0$, recall weighted $4\times$), subject to a precision floor.
- Backtest acts on the signal with a **1-week lag**
 - no same-week look-ahead.

No shuffling, no leakage – numbers are deliberately conservative and deployable.

Metrics for the risk officer

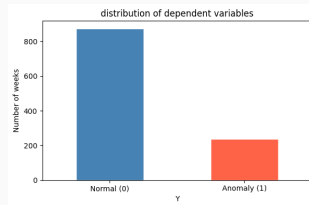
- **Recall / FNR** – a missed crisis is the costly failure.
- **PR-AUC** – honest under heavy class imbalance.
- Precision, F1, ROC-AUC reported for completeness.

Part 1 · Isolation Forest: the warning signal in action



Abnormal periods over time

- Weekly data collection.
- Binary target variable: $Y = 1$ for abnormal periods and $Y = 0$ otherwise.
- Strong class imbalance: anomalies are rare but critical to detect.



Class distribution

2. Isolation Forest: Feature engineering

Feature engineering consists in creating new variables from the original financial time series. The goal is to extract more informative signals for anomaly detection.

Why create new features?

- New variables help capture hidden market dynamics.
- They make the model more sensitive to abnormal periods.

Examples of engineered features

- **Volatility:** measures the instability of a variable.
- **Momentum:** captures short-term upward or downward trends.
- **Drawdown:** detects strong decreases from previous peaks.

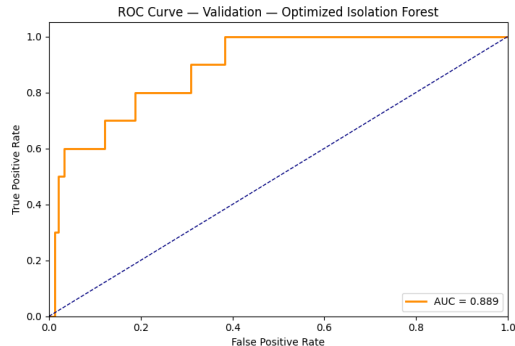
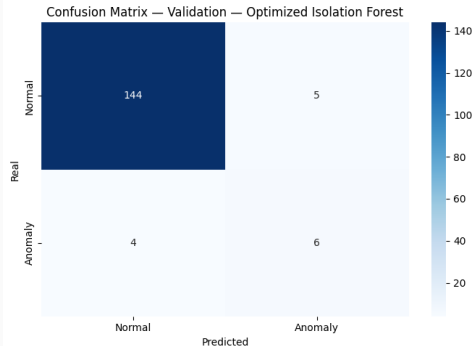
What does it improve?

- Volatility features help identify periods of financial stress.
- Momentum features capture changes in market direction.
- Drawdown features highlight sudden losses or regime shifts.

Feature selection

- We keep variables that bring useful information.
- Highly redundant variables are removed.
- Selected features have low covariance or weak dependence, reducing overlap between signals.

3. Isolation Forest model



Training strategy

- Trained on the training set.
- Threshold optimized on validation data.
- Final evaluation on the test set.

Main idea

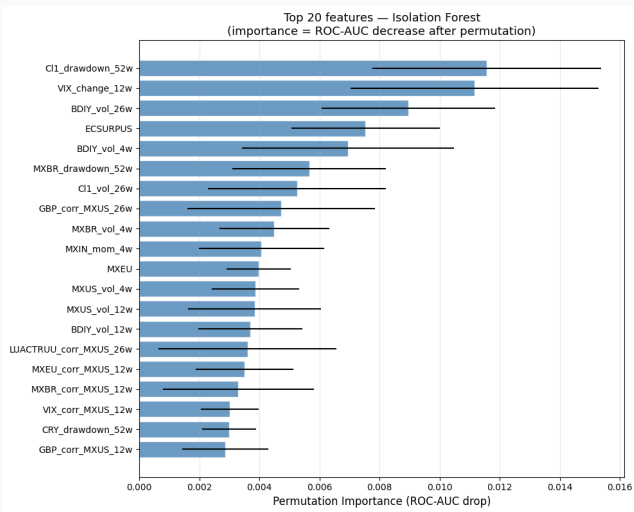
- Anomalies are easier to isolate.
- Fewer splits indicate suspicious points.
- The model outputs an anomaly score.

4. Isolation Forest: Temporal visualization of warning signals



- Crisis periods are associated with a deterioration of the anomaly score.
- The threshold converts a continuous score into an actionable warning signal.
- The threshold choice controls the trade-off between detected crises and false alerts.

5. Isolation Forest: Interpretation and improvement



Feature importance analysis

Key interpretation

- The most important features are consistent with financial stress signals.
- They include oil, VIX, Baltic Dry Index, macroeconomic surprises and drawdowns.
- This improves the interpretability of the anomaly detection model.

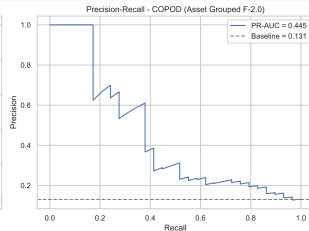
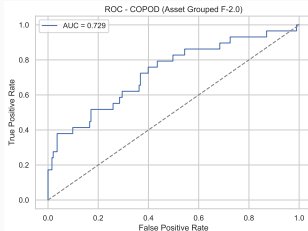
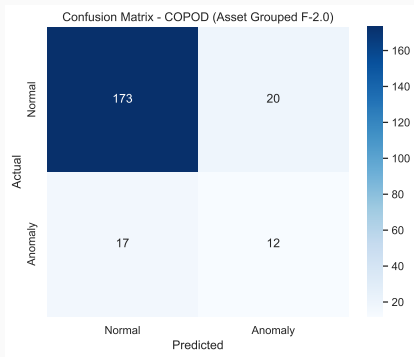
Top-K variant

- More sensitive to anomalies.
- Higher crisis detection rate.
- Also produces more false alerts.

0.76

test recall Top-K

Part 2 · COPOD: copula tails, asset-grouped



Per-feature **min(left, right) tail probability**, aggregated as $-\frac{1}{d} \sum \ln(\cdot)$. Four asset families $\rightarrow$ sub-COPOD $\rightarrow$ meta-COPOD avoids *feature dilution*. Test: recall 41%, PR-AUC 0.45 ($\sim 4\times$ baseline).

Part 2 · COPOD: amber / red operational gates

GREEN – normal regime

190 weeks

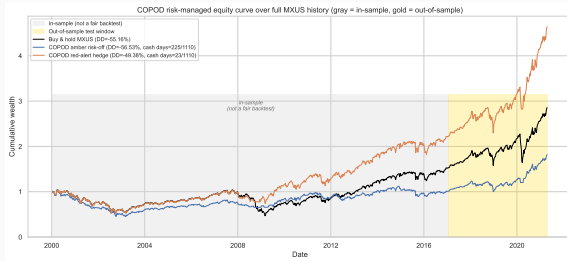
AMBER – monitor (train-normal p80)

27 weeks · 7 crises flagged

RED – hedge (train-normal p99)

5 weeks · 5 crises flagged

12/29 crises caught, 17 missed; 20/20
low-severity false alarms parked in the zero-cost
amber bucket.



Amber rotates to cash – out-of-sample wealth 1.80, DD –18.4%.

Cross-model scoreboard (out-of-sample test)

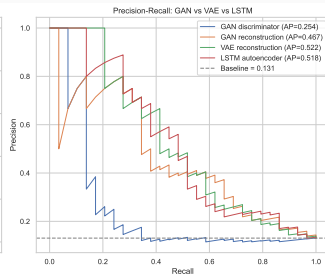
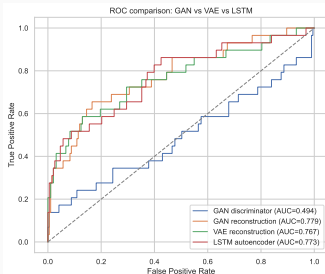
Detector	Caught	Recall	Prec.	F2
Top- K Isolation Forest	24/29	82.8%	58.5%	0.764
Optimized Isol. Forest	14/29	48.3%	58.3%	0.500
COPOD (asset-grouped)	12/29	41.4%	37.5%	0.405
VAE reconstruction	8/29	27.6%	80.0%	0.317
LSTM autoencoder	8/29	27.6%	80.0%	0.317
GAN reconstruction	8/29	27.6%	80.0%	0.317
GAN discriminator	4/29	13.8%	50.0%	0.161

29 true crises in the held-out 2017–2021 test window.

The read

- **No free lunch:** recall vs. precision is a choice, not a ranking.
- **Top- K IF** = recall champion: catches 24 of 29 crises.
- **Deep models** = precision champions: fire rarely, right 80% of the time.
- **COPOD** sits in between.

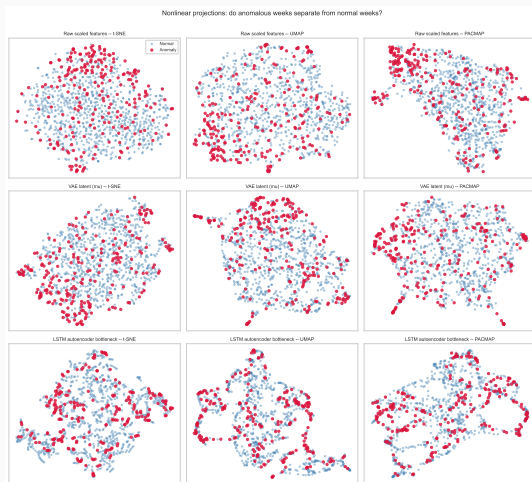
Deep detectors: precise, but cautious



Under the no-leakage split the three reconstruction detectors **converge**: each catches 8/29 crises at 80% precision.

- Best PR-AUC: **VAE 0.52**, LSTM 0.52, GAN-recon 0.47.
- GAN *discriminator* alone is weak (ROC-AUC ≈ 0.49).
- Reconstruction beats realism scoring for novelty.

Does the structure actually separate?

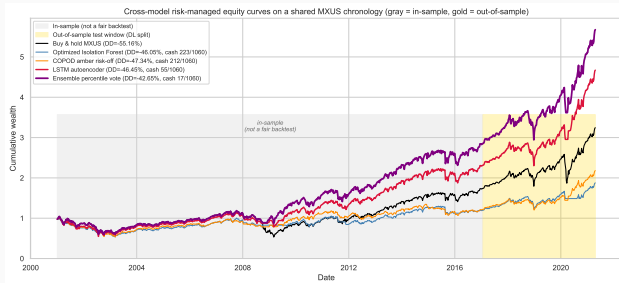


t-SNE / UMAP / PACMAP
on three representations:

- **Raw features** – partial separation only.
- **VAE latent** – barely better.
- **LSTM bottleneck** – clearest cluster; temporal context helps.

Visual proof that adding short-term context sharpens the manifold.

Portfolio impact: every flag cuts drawdown



Test-window backtest

cash when flagged, else MXUS

Strategy	Wealth	Max DD
Buy & hold MXUS	1.87	-27.7%
LSTM autoencoder	2.03	-20.2%
Optimized IF	1.57	-19.7%
COPOD amber	1.80	-18.4%

Signal acts with a 1-week lag (no same-week look-ahead).

- All three **cut drawdown** vs buy & hold ($-27.7\% \rightarrow -18$ to -20%).
- **Only the LSTM** also **grows wealth** above buy & hold; IF/COPOD trade return for deeper drawdown control.

Takeaways

Crises caught

24/29

Top-K Isolation Forest

Best precision

80%

VAE / LSTM / GAN-recon

Beats B&H wealth

2.03

LSTM autoencoder

Min. drawdown

−18.4%

COPOD, vs. −27.7% B&H

- **A strict chronological split** kills look-ahead bias – the numbers are conservative but deployable.
- **No single winner:** pick the detector by mandate. Capital preservation → high-recall Top-K IF; avoid false alarms → high-precision deep models; balanced → COPOD.
- **The robust result:** *every* warning system reduces out-of-sample drawdown (−27.7% → −18 to −20%); the LSTM is the one that also beats buy-and-hold on final wealth.

Bottom line

Framing market stress as novelty detection turns five very different models into one deployable risk-off signal – and the portfolio drawdown reduction survives an honest, leakage-free backtest.